

Defect Detection in Thick CFRP Components

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1.0 Introduction

With the new aerofoil structure our company is bidding for, quality assurance is needed to ensure it meets the required standard when it is delivered from our company. This report discusses the proposed non-destructive testing (NDT) techniques that can be used for the inspection of the solid CFRP component and details how the system will operate. The inspection objectives for each component are to be able to:

- detect delamination larger than 5 mm in diameter anywhere in the component
- verify that average porosity does not exceed 3% through the thickness at any section
- detect disbonding larger than 5 mm in diameter of the adhesive joint between the CFRP and the bonded steel shaft

Based on the objectives stated, the suggested techniques that will be used for inspection are X-ray tomography (XCT) as well as phased-array ultrasonic testing (PAUT), with the overall steps as seen in Figure 1. A discussion on why and how these methods are applied will be presented in the following section.

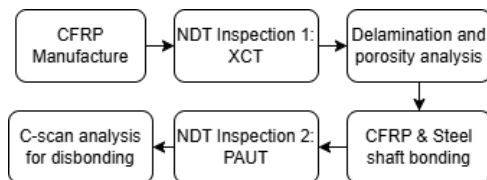


Figure 1. Flow diagram of the overall NDT inspection process

2.0 Methodology

The CFRP component inspected, seen in Figure 2, is a thick laminated composites with a through-thickness value ranging from 37.5-100 mm. This rules out low-frequency NDT such as tap test and mechanical impedance (1). Eddy current was also excluded because it's better suited for MMC rather than PMC, where delamination and porosity are hard to detect due to polymer's lack of electricity conductivity (1). Microwave NDT is also unsuitable because it requires dielectric material, whereas carbon-based composites are conductive. Infrared thermography is eliminated because of its defect detection capability limit of about 5 mm through-thickness (1,2) and rapid dissipation

of heat of big components like ours will lead to the unsuccessful inspection (1). 2D radiography is also not feasible because of its poor detection ability of interlaminar defects (3). This leaves ultrasonic testing (UT) and XCT.

XCT will be used to inspect the overall component for delamination and porosity prior to bonding with the steel shaft. Considering the high through-thickness value of the component as well as its complex geometry, XCT was chosen because of its capability to inspect the whole volumetric interior of an object, providing quantitative 3D information on flaw size, shapes and distribution (4). This inspection must be done before steel shaft bonding to avoid 'metal artifacts', which can cause beam hardening, distracting and obscuring results due to the high-density characteristic of metal (4)

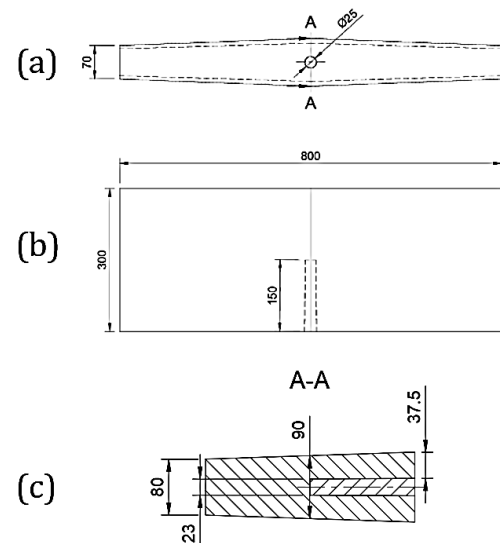


Figure 2. CFRP aerofoil component (a) front, (b) top, and (c) cross-section view

After steel shaft bonding, inspection of the adhesive joint will then be carried out using the PAUT method to detect any disbonding. UT inspection is chosen because of its great capability in detecting defects that are normal to the direction of wave propagation (5) such as delamination/disbonding. In addition, CFRP is anisotropic and highly attenuative, thereby requiring its propagation mode to be a longitudinal wave (5). Pulse-echo ultrasonic method, like PAUT, is used on account of the limited one-way access of the CFRP-steel shaft joint area. The contrast acoustic properties of CFRP, adhesive and steel results in

clear reflection and attenuation differences, allowing disbond to be identified through amplitude and signal comparison using the pulse-echo UT method (6).

Lastly, PAUT is preferred over conventional single-element UT for process efficiency, enabling inspection coverage of larger areas within the required 20-hour production rate (i.e. takt time). Beam steering and electronic focusing capabilities of multiple elements also allow effective inspection of restricted and complex geometry component (2). Immersion and water-jet UT methods are eliminated due to their time-consuming and costly setups, which is not suitable to be done easily within the production line. PAUT offers more flexibility and can be adjusted to the CFRP-steel shaft subassembly set-up.

3.0 Proposed Setup

3.1 NDT Stage I: X-ray Tomography (XCT)

3.1.1 Background

The objective of XCT is to quantify delaminations larger than 5 mm in diameter and verify that the average porosity stays below 3% within the aerofoil.

After evaluating all options, the Nikon VOXLS 40C 450 was found to satisfy the requirements with its dual energy setup of 450kV and 225kV rotating target X-ray sources. High-voltage is necessary as it allows penetration through the hundreds of plies by ensuring a minimum of 10% transmission through the material (4). At 450kV through 100mm of CFRP, the transmission was approximately calculated as 24.4%, satisfying the requirements as shown in Appendix A.

As reflected in previous industry case studies, the 450kV system has proven performance when identifying defects in thick CFRP components (7). Complemented by the 225kV system, Nikon demonstrated that dual energy scanning allows enhanced resolution up to 50 μ m, which will allow us to identify critical delamination and porosities. It clarifies material boundaries that would have otherwise been smeared across multiple voxels (4).

However, while resolution (voxel size) is critical, the geometric sharpness parameter has been raised to be more important. This parameter is controlled by multiple variables, including filter type, orientation, positioning, focal spot size, as well as the ratio of the aerofoil's distance between the source and the detector, as shown in Figure 3 (4).

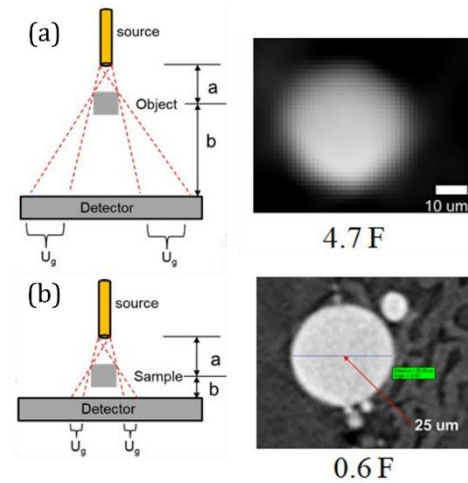


Figure 3. Geometric unsharpness, U_g is defined as $U_g = F \times \frac{b}{a}$. Ensuring the ratio of the object-detector distance, b to the source-object distance, a , has been shown to be more critical than the focal size, F . Modified from (4).

3.1.2 Procedure

Setup of the XCT machine will require optimising for the parameters and calibration, such as attenuation checks. This will require using a calibration block as defined in ISO 19232-5:2018, to verify the minimum voxel size can be achieved at the desired voltage rating (8). Upon calibration, the aerofoil should be set up in the XCT machine as seen in Figure 4.

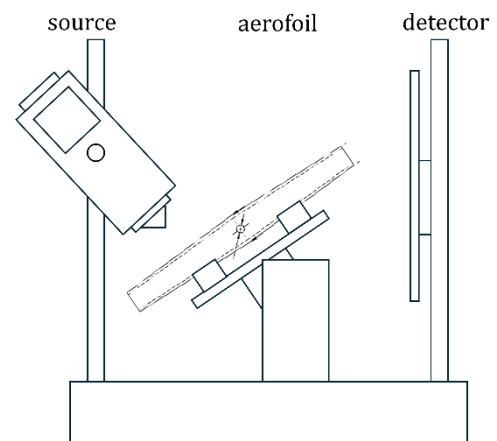


Figure 4. Setup aerofoil at approximately 45° angle to allow closer positioning to the source to minimize energy loss and higher magnification.

It is important to note that high-power XCT machines bring the risk of heavy equipment handling and radiation exposure, as such, a full risk assessment should be conducted prior to operating. After conducting the scan, CT data should be reconstructed using VGStudio MAX software (Hexagon AB, Sweden), here the void content and delamination size can be measured by manipulating the model. VGStudio MAX provides the ability to either automatically calculate the porosity using the VGEasyPore

algorithm, or manually by thresholding specific parameters such as diameter range, sphericity range, and voxel count (9). Porosity, ϕ is calculated by using the formula $\phi = \frac{n_{\text{porosity}}}{n_{\text{total volume}}}$, where n denotes the voxel number (4). For delamination, this can be done by inspecting the cross-section slices manually or using automated solution like VGTrainer which uses machine learning (10).

3.2 NDT Stage II: Phase Array Ultrasonic Testing (PAUT)

3.2.2 Background

To detect the disbands larger than 5mm in the adhesive joint between the CFRP and bonded steel shaft, a PAUT system with a wheeled water-wedge probe was selected. PAUT allows electronic focusing and scanning at multiple angles, ensuring the CFRP-steel interface at a depth of 37.5mm can be interrogated effectively. Additionally, Full Matrix Capture (FMC) and Total Focusing Method (TFM) can be utilised to produce a fully focused image at every point.

A 3.5MHz, 64-element linear phased-array probe, was chosen with studies showing its ability to detect interfacial discontinuities within CFRP joints, supporting its suitability in detecting 5mm disbands (11). This provides a balance between resolution and penetration of the CFRP, whereby higher frequencies may potentially experience severe attenuation. Higher frequencies will not be considered by reason of growth of attenuation.

Considerable studies have been conducted demonstrated the ability to detect defects up to 74mm thick in GFRP using a 0.5MHz linear-array probe (12). However, due to the lack of inspection method investigation of CFRP of this thickness, it is recommended to begin with the 3.5 MHz frequency probe and if disbands are not detectable, consider a lower frequency of 1 MHz probe to achieve sufficient penetration. In particular, the RollerFORM phased-array wheel probe by Evident Scientific is selected for its complete scanning system, ability to detect defects in CFRP up to 50mm thick, built-in laser guide, and continuous water coupling (13). Hence, eliminating the need for any full immersion testing or robotic arm scanning over the aerofoil. The OmniScan X4 unit is the phased-array instrument selected due to its FMC/TFM capability, C-scan generation, and compatibility with both the 3.5 MHz and 1 MHz RollerFORM probes (14).

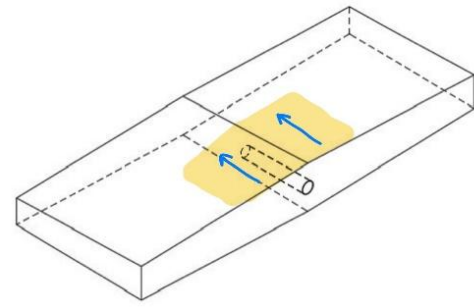


Figure 5. Direction of scanning path using the RollerFORM.

3.2.2 Procedure

Upon acquiring the equipment necessary, surface preparation should be done on the CFRP surface to remove any dust, grease, or debris. This ensures optimal ultrasonic coupling when the water couplant is applied via the RollerFORM. The beam focus will be set at the bondline using the system's built-in laser guide, with a delay line height of 25mm and elevation of 6.4mm in accordance with the model's parameters, optimising the beam path and ensure straight precise scans. Scanning will be performed along the length of the steel shaft near the centre of the aerofoil on both the upper and lower side as shown in Figure 5. The phased-array probe should operate with a conventional delay-and-sum (DAS) beamforming technique which applies specific time delays across each transducer element, enabling multiple angles of inspection to be conducted without having to reposition the probe. Known CFRP and steel shaft thickness should be used to configure focal depth, time-of-flight (TOF), and steering parameters. A calibration block with ISO 18563-1 will be used to verify sensitivity, TOF, and focal depth to ensure that 5mm disbands at the CFRP-steel interface can be accurately detected (15).

The RollerFORM scanner's encoder is zeroed and checked for positional accuracy, and the water coupling is verified to be stable along the scan path. Scanning is performed in C-scan mode whilst maintaining steady speed and consistent contact to generate high-resolution 2D images of the bondline which will be used for subsequent analysis. C-scan is the most efficient method which provides a top-down representation of the signal amplitude and TOF maps of the bondline, making it the most reliable option to detect disbands (5). The OmniScan X4 reconstructs the data into a C-scan image, allowing measurement of the size, shape, and spatial coordinates of any detected disbands. FMC acquisition and TFM processing can be implemented to improve the detectability of disbands and improve resolution where necessary. Disbands in the adhesive joints are therefore

identified with PAUT interrogating the interface between the CFRP and steel, identifying any changes in interface echoes, bondline reflection amplitude, phase shifts, and altered TOF (16). By analysing the amplitude of ultrasonic reflections at the bondline, C-scan would then convert the amplitude into a 2D top-down map of the adhesive layer where the bonded regions appear as high-amplitude areas while the disbanded regions as low-amplitude areas, corresponding to the ultrasonic wave being completely reflected at the first interface and indicating the presence of a disbond in the adhesive layer (17).

4.0 Limitations

While effective, significant costs come with purchasing and maintaining XCT machines. In addition, the large file sizes, scan times and limited scan volume are constraints which limit its versatility and scalability, requiring careful production planning. There is the risk of the lower detail captured as using a voxel size of $50\mu\text{m}$ may miss small porosities due to the requirement of 3-5 times voxels across the feature for detection (4). Larger voxels may also overestimate the actual porosity, which can be partially addressed by implementing deep learning and careful thresholding segmentation during post-processing (18).

The inspection of the CFRP-steel interface is a complex task due to the difference of acoustic impedances of the materials which will have a strong reflection even without the presence of defects. Phased-array probes may struggle to distinguish disbonds when operating at 3.5MHz due to the small amplitude difference of interface and defect reflections (6). Lowering to a small frequency of 1MHz as mentioned previously may resolve this. Another way to improve the probability of detecting the disbonds would be to examine subsequent multiple reflections which would produce a larger difference in amplitude between the good and defected areas, therefore providing a more reliable basis for disbond detection (6).

Summary

We have proposed an NDT strategy for inspecting the thick CFRP aerofoil component via XCT and PAUT. Using a dual-energy 450/225kV setup, XCT provides full volumetric inspection and quantification of porosity and delamination before bonding to avoid metal artifacts. After bonding, PAUT is used to evaluate the adhesive joint, employing a RollerFORM phased-array probe with FMC/TFM capability to overcome access restrictions and material attenuation. Limitations relate to the high cost of XCT machines, impedance mismatch at the CFRP-steel interface and

potential signal interpretation challenges should be noted during the PAUT inspection so that necessary in-process frequency and post-processing adjustments can be made.

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Appendix A

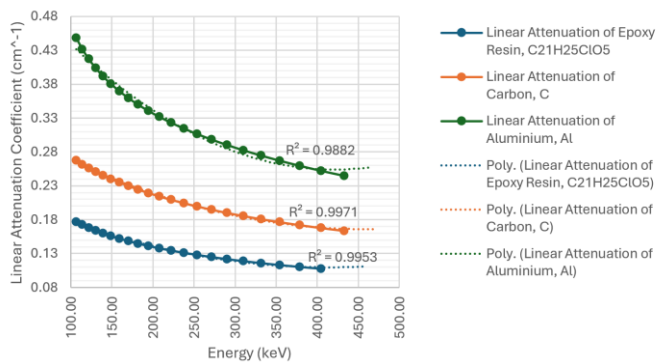
Assuming a fibre volume fraction of 60%, carbon fibre (PAN precursor) density of 1.81 g/cm^3 , and epoxy resin density of 1.10 g/cm^3 , we can calculate a final density of 1.526 g/cm^3 . Given a volume of $1,830 \text{ cm}^3$, the total mass can be calculated to be 27.9 kg .

The transmission percentage, $\frac{I}{I_0}$ is calculated using the Beer-Lambert law with linear attenuation coefficient, μ and thickness, t . Linear attenuation coefficient can be calculated using the mass attenuation coefficient, which was calculated using the individual densities of each material.

Assumptions made:

- fibre volume fraction of 60%
- CFRP material behaves like a mixture, therefore the attenuation contribution of each material can be added by their volume fraction, equating to 0.14097 cm^{-1}
- The thickness of the material is 100 mm .

$$\frac{I}{I_0} = e^{-\mu t} = e^{-(0.14097)(10)} = 24.42\%$$



Note: Linear Attenuation Coefficient (cm⁻¹) varies based on the Energy (keV) of epoxy resin and carbon fibre (carbon). Aluminium for comparison was shown here. Despite being lightweight, the attenuation is much higher than carbon and epoxy resin.